Alex Fernsby

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Profile

Data engineer with 7 years of experience building batch and streaming pipelines for retail and logistics companies. Owns shipment-event processing end to end; mentors two junior engineers and leads the weekly data-modelling review.

Core Competencies

- **Pipeline engineering**: Python, SQL, Apache Airflow, dbt and Kafka.
- **Data warehouse**: PostgreSQL and BigQuery; partitioning and clustering.
- **Deployments**: Terraform and GitHub Actions; Docker.
- **Data quality**: Great Expectations tests and pipeline alerting.
- **Collaboration**: data contracts, documentation and mentoring.

Professional Experience

Tidewater Logistics **Copenhagen**
Senior Data Engineer *March 2022 - present*

- Own the shipment-events pipeline end to end in logistics, including real-time Kafka intake: 40 million events per day from Kafka into a PostgreSQL and BigQuery data warehouse, orchestrated with Apache Airflow.
- Deliver lower latency for batch data pipelines: cut the nightly delivery-performance mart rebuild from 6 hours to 55 minutes with incremental dbt models.
- Mentor two junior engineers and lead the weekly data-modelling review.
- Introduced data-quality tests with Great Expectations on 30 core tables; failed loads dropped 70% in the first quarter.
- Moved the team's deployments to GitHub Actions for CI/CD and Terraform; new environments take 20 minutes instead of two days.
- Defined technical requirements in the shipment-events data contract with five producing teams, supporting data governance; wrote the pipeline's on-call runbook.
- Cut the BigQuery events warehouse bill by 35% through partitioning and clustering the five largest tables.
- Built a Looker explore so operations planners answer delivery-delay questions themselves instead of filing tickets.
- Ran interviews for three data-engineering hires and wrote the team's take-home exercise.

Marlow Retail Group **Utrecht**
Data Engineer *June 2018 - February 2022*

- Built the company's first Airflow deployment and migrated 45 cron jobs onto it.
- Designed a customer-orders star schema in PostgreSQL used by 12 analysts.
- Wrote Python ingestion services for partner price feeds over REST and SFTP, handling about 2 million rows a day.
- Supported the warehouse migration from on-premise PostgreSQL to BigQuery, validating 60 tables row by row.
- Trained 12 analysts on dbt and reviewed their first 40 models.

- Added alerting on 45 Airflow jobs so late partner feeds are noticed within 15 minutes instead of the next morning.
- Wrote the migration guide for moving reports off the old warehouse, used by four teams.

Delft Energy Cooperative **Delft**
Junior Analyst *September 2016 - May 2018*

- Built automated data solutions with Python and SQL for monthly consumption reports, saving about 3 days each month.
- Built a small Flask dashboard used by 25 cooperative members to read their consumption.
- Cleaned five years of meter readings with Pandas and documented the resulting 14 data-quality rules.
- Presented consumption findings to the cooperative's board twice a year.

Projects

Open-source dbt package for freight-rate data tests, 120 GitHub stars: freight-tests.

2024: "Data contracts in practice", Rotterdam Data Meetup talk.

Certifications

2023: Google Cloud Professional Data Engineer

2022: dbt Analytics Engineering Certification

Education

Delft University of Technology
MSc Computer Science (data systems) *2016*

Delft University of Technology
BSc Computer Science *2014*

Languages

English: C1; Dutch: native; German: A2.

Publications

2024: "Testing freight data with dbt", Data Engineering Weekly guest post.

Honors and Awards

2023: Tidewater Logistics Engineering Excellence Award